

SPLUNK HOME LAB
PROJECT REPORT
BY
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ABSTRACT

This project demonstrates the creation of a Home SOC (Security Operations Center) Lab using Splunk Enterprise and Kali Linux. The objective was to monitor system and authentication logs, simulate brute-force attacks, and analyze security events in a centralized SIEM environment. The project helped in understanding SIEM concepts, log analysis, threat detection, and SOC monitoring in a safe virtualized environment.

INTRODUCTION

In today's rapidly evolving cybersecurity landscape, organizations face increasingly sophisticated threats, necessitating the need for robust Security Operations Centers (SOCs). A SOC is crucial for detecting, analyzing, and responding to security incidents to safeguard critical assets and data. However, the process of building, configuring, and maintaining a SOC can be resource intensive, often requiring significant expertise and infrastructure.

Security monitoring and centralized log management are essential components of modern cybersecurity operations. In this project, Splunk Enterprise was configured on a Windows machine while Kali Linux acted as the log-generating endpoint. Splunk Universal Forwarder was installed on Kali Linux to forward system logs to Splunk Enterprise for analysis and monitoring.

REQUIREMENTS

Hardware & Software Requirements:

- Windows System with Splunk Enterprise
- Kali Linux Virtual Machine
- Splunk Universal Forwarder
- VirtualBox / VMware
- Network connectivity between systems
- Windows Firewall configuration for port 9997

Splunk – It is a lightweight agent developed by Splunk that collects and forwards logs and machine data from endpoints to a Splunk Enterprise server for centralized monitoring and analysis.

It is commonly used to send:

- System logs
- Authentication logs
- Security events
- Application logs
- Network logs

In this project, the Splunk Universal Forwarder was installed on Kali Linux to forward **authentication logs** and **system logs** to the Windows-based Splunk Enterprise server for real-time security monitoring and threat detection.

SETUP & CONFIGURATION

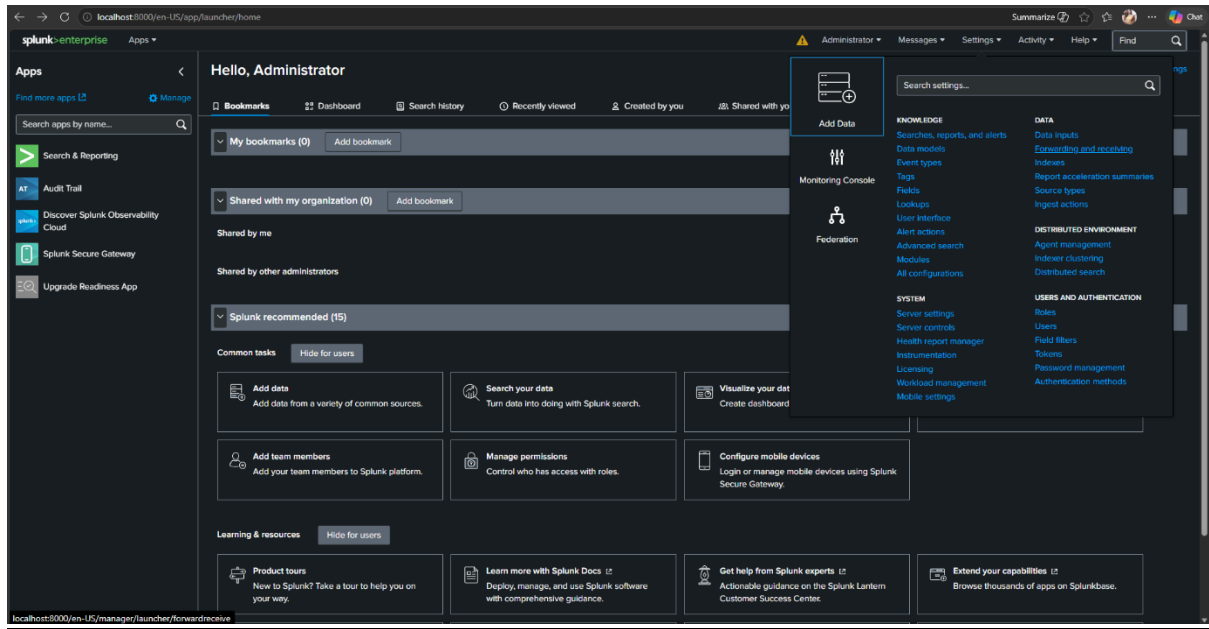
1. Installed Splunk Enterprise on Windows.
2. Configured receiving port 9997 in Splunk.
3. Allowed inbound firewall rule for port 9997.
4. Installed Splunk Universal Forwarder on Kali Linux.
5. Added Windows Splunk server as the forward server.
6. Monitored authentication and system logs from /var/log.
7. Verified data ingestion in Splunk Search & Reporting.

BRUTE FORCE ATTACK SIMULATION

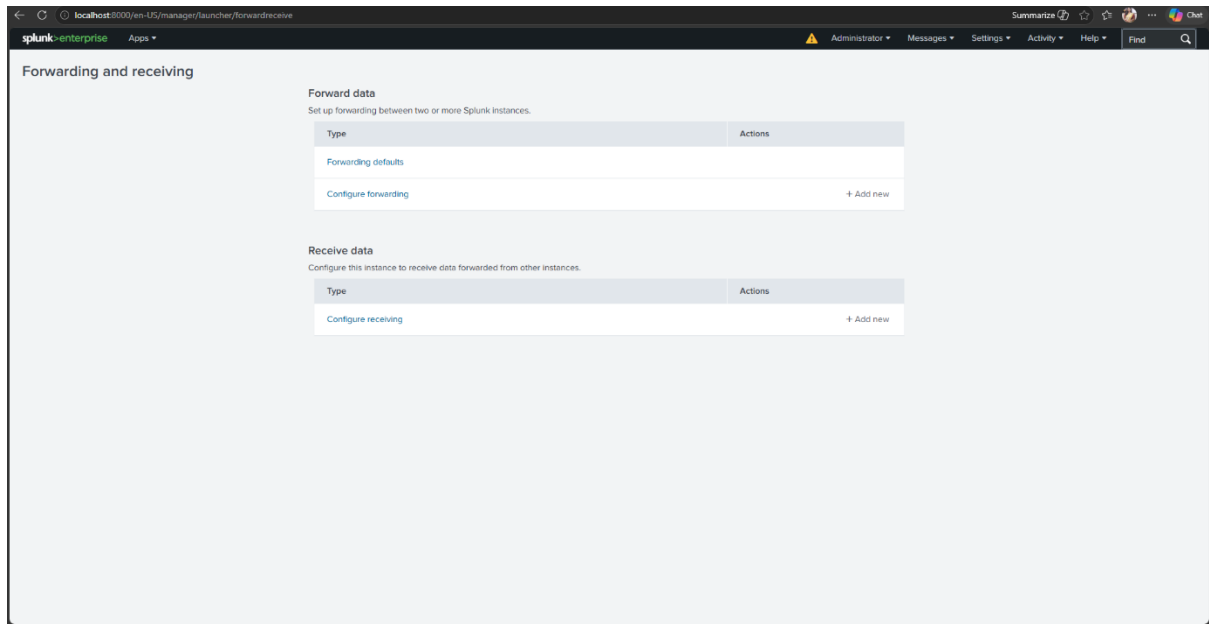
A controlled brute-force login attempt was performed using SSH from the Windows system to Kali Linux. Failed authentication attempts were successfully captured in /var/log/auth.log and forwarded to Splunk Enterprise. Splunk Search Processing Language (SPL) queries were used to identify failed password attempts and suspicious login behavior.

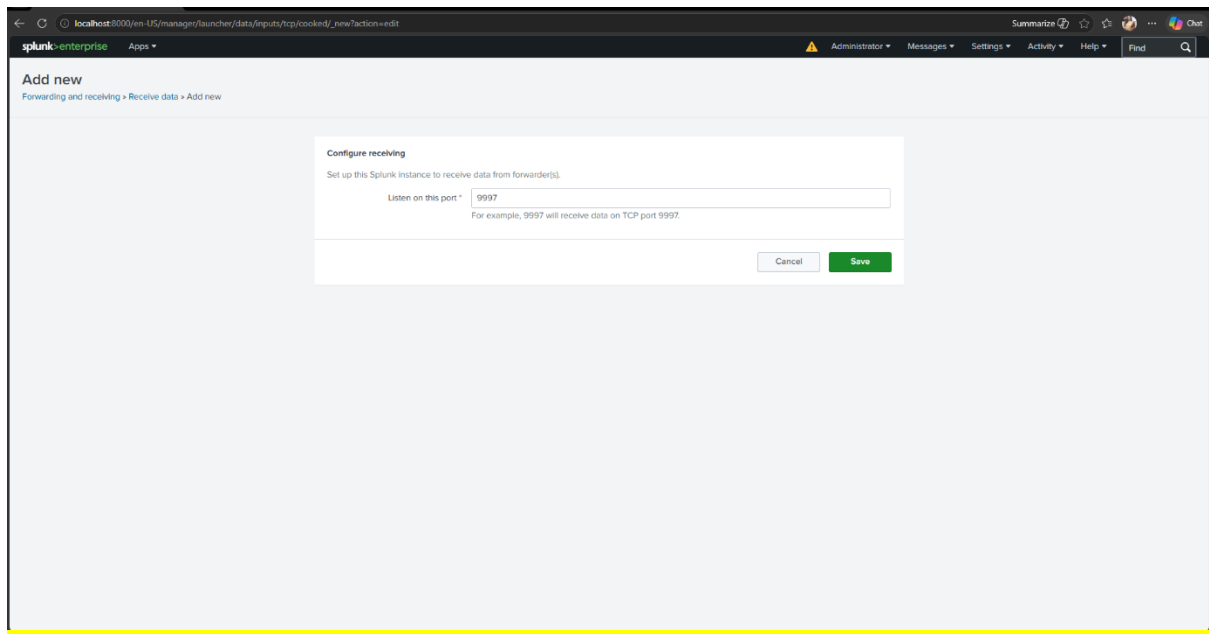
SCREENSHOTS

Splunk Dashboard

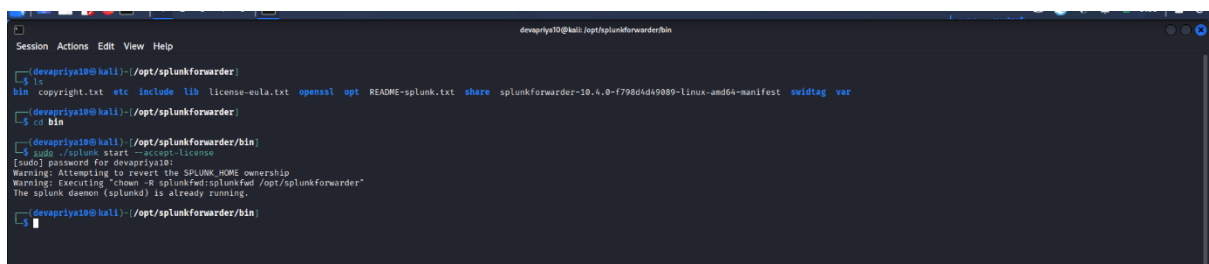


Setting up the port





Installed Splunk forwarder in Kali Linux and Performed actions for setup.



Windows ip address.

```
Command Prompt
Connection-specific DNS Suffix . : 
Link-local IPv6 Address . . . . . : fe80::...
IPv4 Address. . . . . : 192.168.18.1
Subnet Mask . . . . . : 255.255.255.0
Default Gateway . . . . . : 

Wireless LAN adapter Wi-Fi 2:

Media State . . . . . : Media disconnected
Connection-specific DNS Suffix . : 

Wireless LAN adapter Wi-Fi 5:

Media State . . . . . : Media disconnected
Connection-specific DNS Suffix . : 

Wireless LAN adapter Wi-Fi:

Connection-specific DNS Suffix . : 
Link-local IPv6 Address . . . . . : fe80::...
IPv4 Address. . . . . : 192.168.18.1
Subnet Mask . . . . . : 255.255.255.0
Default Gateway . . . . . : 192.168.18.1

Ethernet adapter Bluetooth Network Connection:

Media State . . . . . : Media disconnected
Connection-specific DNS Suffix . : 

C:\Users\hp>
```

Adding forward-server to splunk [ip:port]

```
(devapriya10@kali)-[/opt/splunkforwarder/bin]
$ sudo ./splunk add forward-server :9997
Warning: Attempting to revert the SPLUNK_HOME ownership
Warning: Executing "chown -R splunkfwd:splunkfwd /opt/splunkforwarder"
Your session is invalid. Please login.
Splunk username: admin
Password:
:9997 forwarded-server already present

(devapriya10@kali)-[/opt/splunkforwarder/bin]
$
```

Finding logs in Linux. Required logs for this project is syslog and auth.log .

```
devapriya10@kali: /opt/splunkforwarder/bin
$ cd /var/log
$ ls
alternatives.log      alternatives.log.4.gz  boot.log      boot.log.4  btmp      dpkg.log.1  dpkg.log.5.gz  installer  lightdm      macchanger.log.3.gz  nginx      private  samba      sysstat      Xorg.0.log
alternatives.log.1    apache2              boot.log.1    boot.log.5  btmap.1   dpkg.log.2.gz  fontconfig.log  journal    macchanger.log  macchanger.log.4.gz  notas-scanner  README  speech-dispatcher  user.log    Xorg.0.log.oid
alternatives.log.2.gz apt                  boot.log.2    boot.log.6  cron.log  dpkg.log.3.gz  gdm             kdm.log    macchanger.log.1.gz  Mosquitto  Openvpn  redis      stunnel4    utmp      Xorg.0.log.oid
alternatives.log.3.gz auth.log             boot.log.3    boot.log.7  dpkg.log  dpkg.log.4.gz  inetd           lastlog    macchanger.log.2.gz  mysql      postgres  runit      syslog      wtmp      wtmp.db

devapriya10@kali: /var/log
$
```

```
devapriya10@kali: /opt/splunkforwarder/bin
$ sudo ./splunk add monitor /var/log/syslog
Warning: Attempting to revert the SPLUNK_HOME ownership
Warning: Executing "chown -R splunkfd:splunkfd /opt/splunkforwarder"
Cannot create another input with the name "/var/log/syslog", one already exists.

devapriya10@kali: /opt/splunkforwarder/bin
$ sudo ./splunk add monitor /var/log/auth.log
Warning: Attempting to revert the SPLUNK_HOME ownership
Warning: Executing "chown -R splunkfd:splunkfd /opt/splunkforwarder"
Cannot create another input with the name "/var/log/auth.log", one already exists.

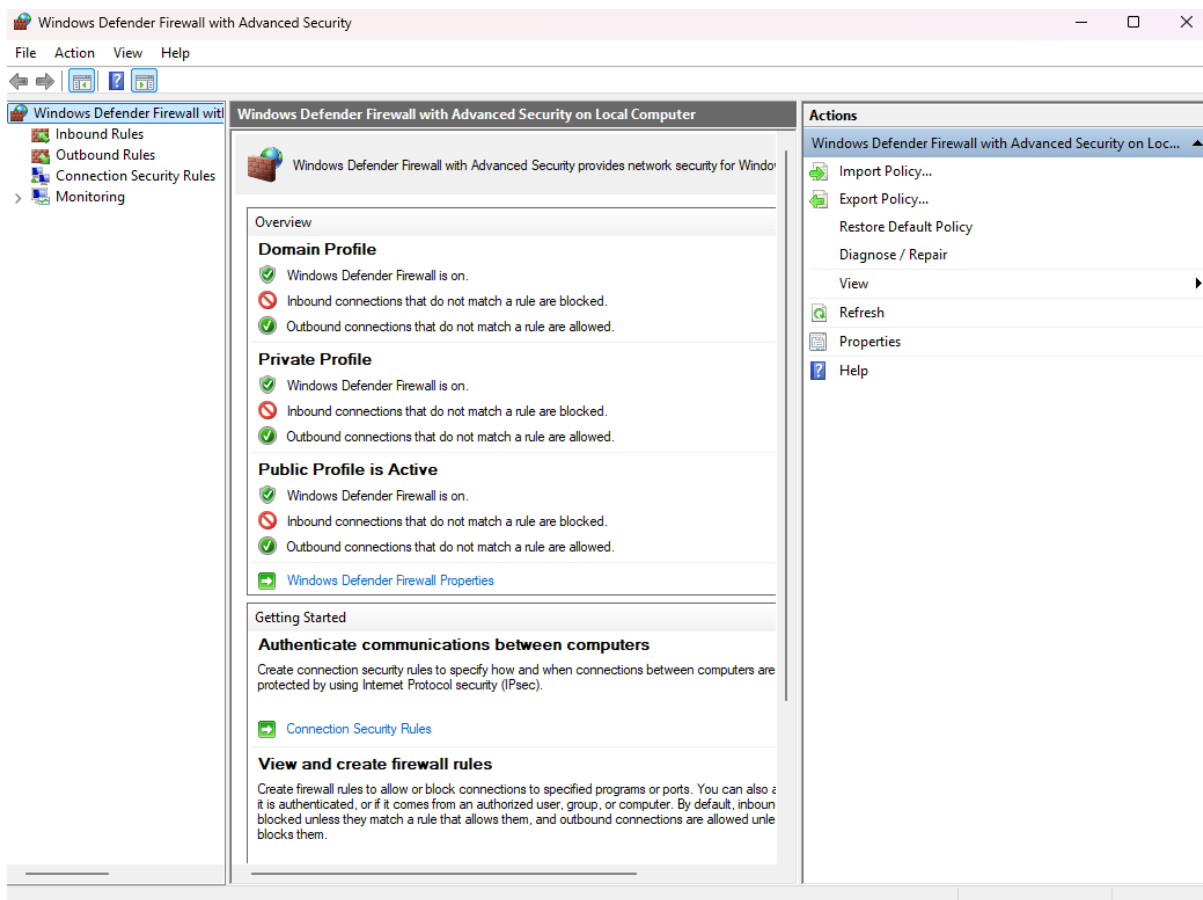
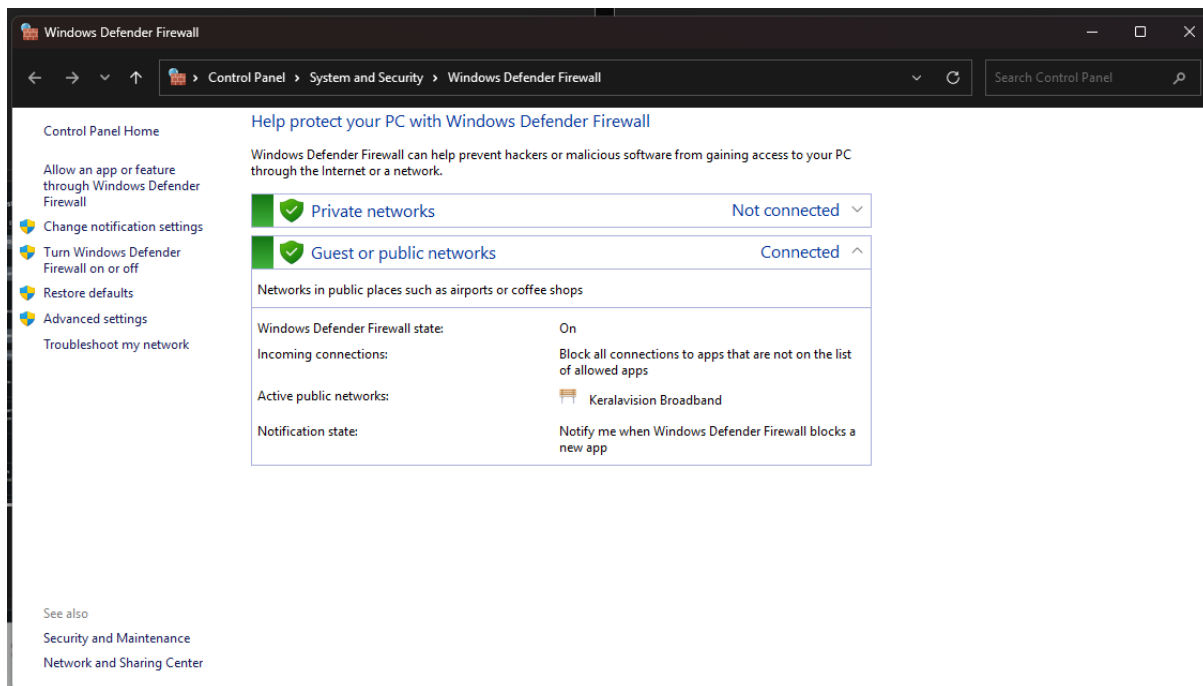
devapriya10@kali: /opt/splunkforwarder/bin
$
```

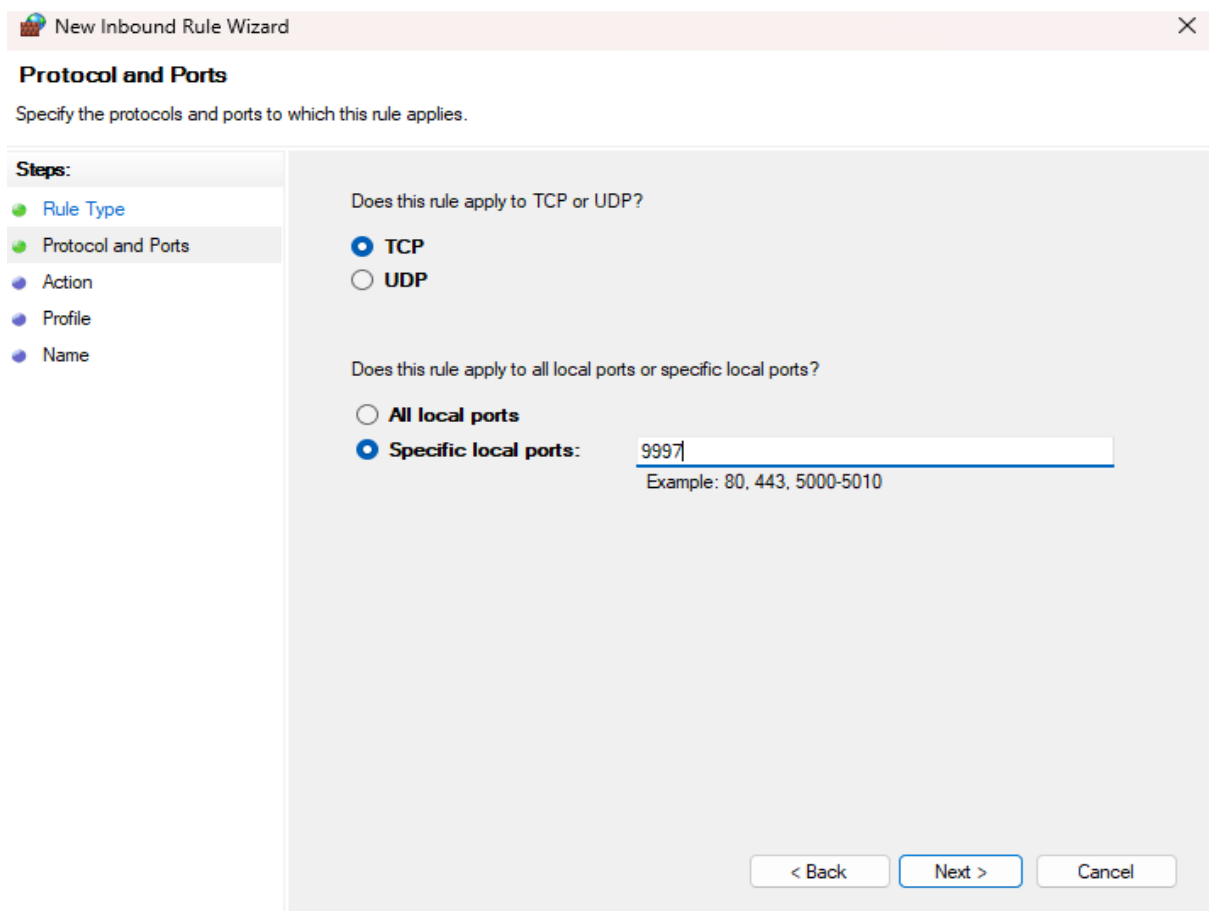
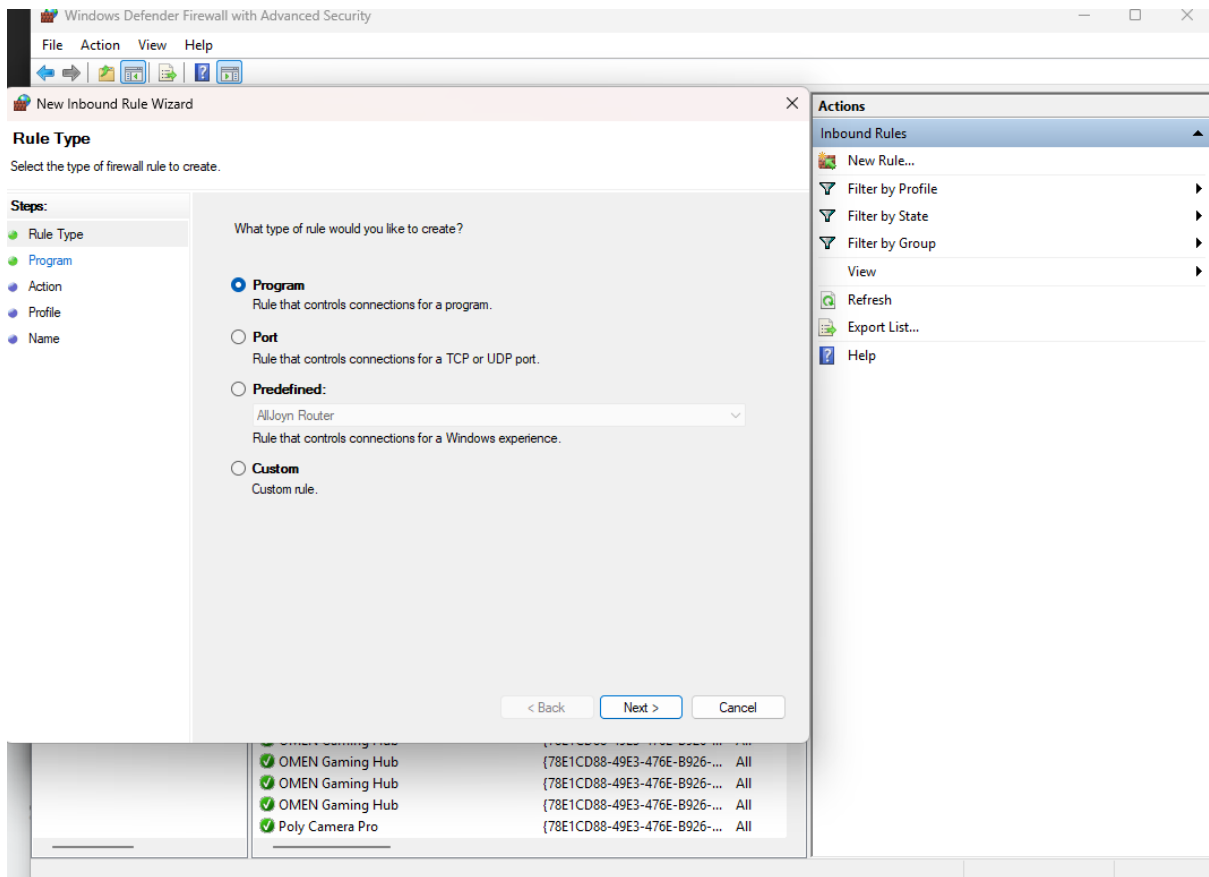
```
devapriya10@kali: /var/log
$ sudo systemctl status ssh
[mode] password for devapriya10:
● ssh.service - OpenSSH Secure Shell server
   Loaded: loaded (/usr/lib/systemd/system/ssh.service; enabled; preset: disabled)
   Active: active (running) since Wed 2020-05-27 09:50:12 IST; 6min ago
   Invocation: 39a311b0fe3647338e9451b8ca8960c
   Docs: man:ssh(8)
        man:ssh_config(5)
   Process: 816 ExecStartPre=/usr/sbin/ssh -t (code=exited, status=0/SUCCESS)
   Main PID: 829 (sshd)
   Tasks: 1 (limit: 9959)
   Memory: 1.0M (peak: 2.4M)
   CPU: 72ms
   CGroup: /system.slice/ssh.service
           └─829 "sshd: /usr/sbin/sshd -D [listener] 0 of 10-100 startups"

May 27 09:50:12 kali systemd[1]: Starting ssh.service - OpenSSH Secure Shell server...
May 27 09:50:12 kali sshd[829]: Server listening on 0.0.0.0 port 22.
May 27 09:50:12 kali systemd[1]: Started ssh.service - OpenSSH Secure Shell server.
May 27 09:50:12 kali sshd[829]: Server listening on :: port 22.

devapriya10@kali: /var/log
$
```

Setting up Firewall Inbound rules.





New Inbound Rule Wizard

Action

Specify the action to be taken when a connection matches the conditions specified in the rule.

Steps:

Rule Type

Protocol and Ports

Action

Profile

Name

What action should be taken when a connection matches the specified conditions?

☒ **Allow the connection**
This includes connections that are protected with IPsec as well as those are not.

☐ **Allow the connection if it is secure**
This includes only connections that have been authenticated by using IPsec. Connections will be secured using the settings in IPsec properties and rules in the Connection Security Rule node.

Customize...

☐ **Block the connection**

< Back

Next >

Cancel

New Inbound Rule Wizard

Profile

Specify the profiles for which this rule applies.

Steps:

Rule Type

Protocol and Ports

Action

Profile

Name

When does this rule apply?

☒ **Domain**
Applies when a computer is connected to its corporate domain.

☒ **Private**
Applies when a computer is connected to a private network location, such as a home or work place.

☒ **Public**
Applies when a computer is connected to a public network location.

< Back

Next >

Cancel

New Inbound Rule Wizard

X

Name

Specify the name and description of this rule.

Steps:

● Rule Type

● Protocol and Ports

● Action

● Profile

● Name

Name:

splunk forwarder

Description (optional):

< Back

Finish

Cancel

✓ splunk forwarder

All

Found Host kali in Splunk. Forwarder is Successful.

The screenshot shows the Splunk Search interface. A 'Data Summary' panel is open, displaying search results for the host 'kali'. The panel includes a table with columns: Host, Count, and Last Update. The results show that the host 'kali' has a count of 3,866 and was last updated on 5/27/26 9:58:28.000 AM.

Host	Count	Last Update
LAPTOP-KK77CQ9S	24144	5/27/26 9:58:22.000 AM
kali	3,866	5/27/26 9:58:28.000 AM

The screenshot shows the Splunk Search interface with a 'New Search' panel. The search query is 'host=kali source="/var/log/auth.log"'. The results show 113 events. The 'Events' tab is selected, displaying a list of events with columns: Time, Event, and Source. The events are sorted by time, showing a sequence of SSH sessions and authentication failures for the user 'devapiya10' on host 'kali'.

Time	Event	Source
5/27/26 10:22:28.493 AM	2026-05-27T18:22:28.493Z host=kali sshd-session[16338]: PAM 2 more authentication failures; logname=uid=0 euid=0 tty=ruser= rhost=192.168.18.33 user=devapiya10	/var/log/auth.log
5/27/26 10:22:28.491 AM	2026-05-27T18:22:28.491Z host=kali sshd-session[16338]: Connection reset by authenticating user devapiya10 192.168.18.33 port 58852 [preauth]	/var/log/auth.log
5/27/26 10:22:27.004 AM	2026-05-27T18:22:27.004Z host=kali sshd-session[16338]: Failed password for devapiya10 from 192.168.18.33 port 58852 ssh2	/var/log/auth.log
5/27/26 10:22:25.191 AM	2026-05-27T18:22:25.191Z host=kali sshd-session[16338]: pam_winbind(sshd:auth): internal module error (retval = PAM_AUTHINFO_UNAVAIL(9), user = "devapiya10")	/var/log/auth.log
5/27/26 10:22:25.191 AM	2026-05-27T18:22:25.191Z host=kali sshd-session[16338]: pam_winbind(sshd:auth): request wbcLogonUser failed: WBC_ERR_WINBIND_NOT_AVAILABLE, PAM error: PAM_AUTHINFO_UNAVAIL (9)	/var/log/auth.log
5/27/26 10:22:25.190 AM	2026-05-27T18:22:25.190Z host=kali sshd-session[16338]: pam_winbind(sshd:auth): pam_get_item returned a password	/var/log/auth.log
5/27/26 10:22:25.190 AM	2026-05-27T18:22:25.190Z host=kali sshd-session[16338]: pam_winbind(sshd:auth): getting password (0x00000388)	/var/log/auth.log
5/27/26 10:22:25.189 AM	2026-05-27T18:22:25.189Z host=kali unix_chkpwd[10462]: password check failed for user (devapiya10)	/var/log/auth.log
5/27/26 10:22:22.885 AM	2026-05-27T18:22:22.885Z host=kali sshd-session[16338]: Failed password for devapiya10 from 192.168.18.33 port 58852 ssh2	/var/log/auth.log

Brute Force Attack simulation , to get logs into splunk .

```
Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

Install the latest PowerShell for new features and improvements! https://aka.ms/PSWindows

PS C:\Users\hp> ssh devapriya10@192.168.18.85
The authenticity of host '192.168.18.85 (192.168.18.85)' can't be established.
ED25519 key fingerprint is SHA256:muN12DpAWaAbEAgjwmCtiqnMYj8ZA0AJLSl6D6WEw+Y.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '192.168.18.85' (ED25519) to the list of known hosts.
devapriya10@192.168.18.85's password:
Permission denied, please try again.
devapriya10@192.168.18.85's password:
Permission denied, please try again.
devapriya10@192.168.18.85's password:
devapriya10@192.168.18.85: Permission denied (publickey,password).
PS C:\Users\hp> |
```

splunk-enterprise Apps

Administrator Messages Settings Activity Help Find

Search Analytics Datasets Reports Alerts Dashboards Modules

Search & Reporting

New Search

Save As Create Table View Close

host="kali" "failed password" source="/var/log/auth.log" Time range: Last 24 hours

3 events (5/26/26 10:30:00.000 AM to 5/27/26 10:56:40.000 AM) No Event Sampling Job

Events (3) Patterns Statistics Visualization

Timeline format Zoom Out Zoom to Selection Deselect 1 hour per column

Format Show: 20 Per Page View: List

	Time	Event
>	5/27/26 10:22:27.004 AM	2626-65-27718:22:27.064387485:38 kali sshd-session[16338]: failed password for devapriya10 from 192.168.18.33 port 58852 ssh2 host = kali source = /var/log/auth.log sourcetype = auth-2
>	5/27/26 10:22:22.885 AM	2626-65-27718:22:22.885897485:38 kali sshd-session[16338]: failed password for devapriya10 from 192.168.18.33 port 58852 ssh2 host = kali source = /var/log/auth.log sourcetype = auth-2
>	5/27/26 10:22:16.765 AM	2626-65-27718:22:16.765889485:38 kali sshd-session[16338]: failed password for devapriya10 from 192.168.18.33 port 58852 ssh2 host = kali source = /var/log/auth.log sourcetype = auth-2

< Hide Fields All Fields

SELECTED FIELDS

- # host 1
- # source 1
- # sourcetype 1

INTERESTING FIELDS

- # date_hour 1
- # date_mday 1
- # date_minute 1
- # date_month 1
- # date_second 3
- # date_wday 1
- # date_year 1
- # date_zone 1
- # index 1
- # linecount 1
- # packet 1
- # splunk_server 1
- # timeendpos 1
- # timenstartpos 1

+ Extract New Fields

KEY LEARNINGS

- Understanding SIEM fundamentals
- Log forwarding and centralized monitoring
- Splunk Search Processing Language (SPL)
- Authentication log analysis
- Brute-force attack detection
- Basic SOC monitoring and investigation

CONCLUSION

This project provided practical hands-on experience in building a basic Home SOC (Security Operations Center) environment using Splunk Enterprise and Kali Linux. Through configuring log forwarding, monitoring authentication events, and analyzing brute-force attack attempts, the project helped in understanding how real-world SOC environments detect and investigate suspicious activities.

The implementation of Splunk Universal Forwarder enabled centralized log collection from Kali Linux to the Splunk Enterprise server, demonstrating the importance of SIEM solutions in cybersecurity operations. Using Splunk Search Processing Language (SPL), different security events and failed login attempts were identified and analyzed effectively.

Overall, this lab environment served as an excellent platform for learning security monitoring, threat analysis, and defensive cybersecurity concepts in a safe and controlled environment. The project also strengthened practical troubleshooting and problem-solving skills related to networking, log forwarding, and security event analysis.